INCH-POUND

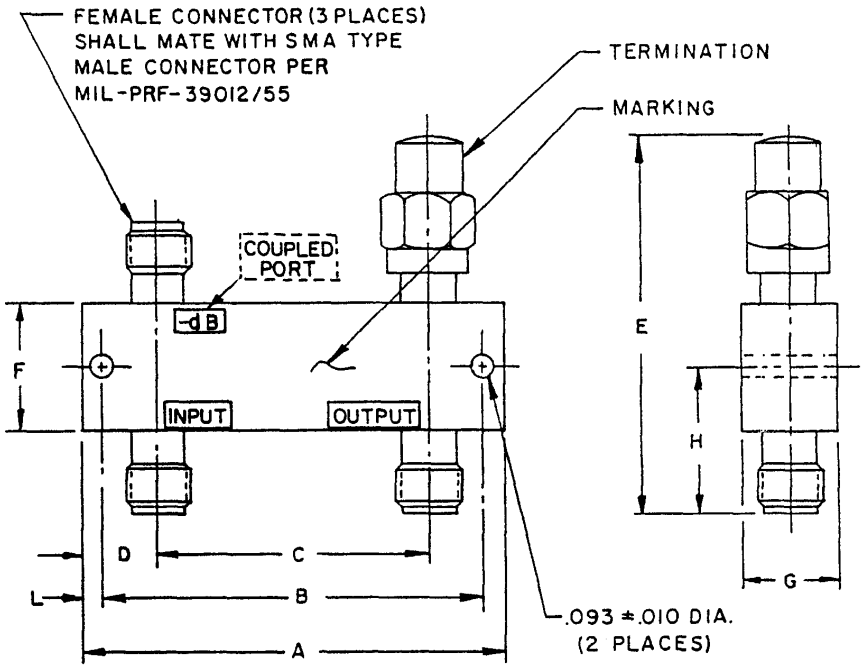
MIL-DTL-15370/10C
18 June 2003
SUPERSEDING
MIL-C-15370/10B
22 September 1989

DETAIL SPECIFICATION SHEET

COUPLERS, DIRECTIONAL (COAXIAL, SMA CONNECTORS) (.50 to 18 GHz)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the couplers described herein shall consist of this specification sheet and MIL-DTL-15370.

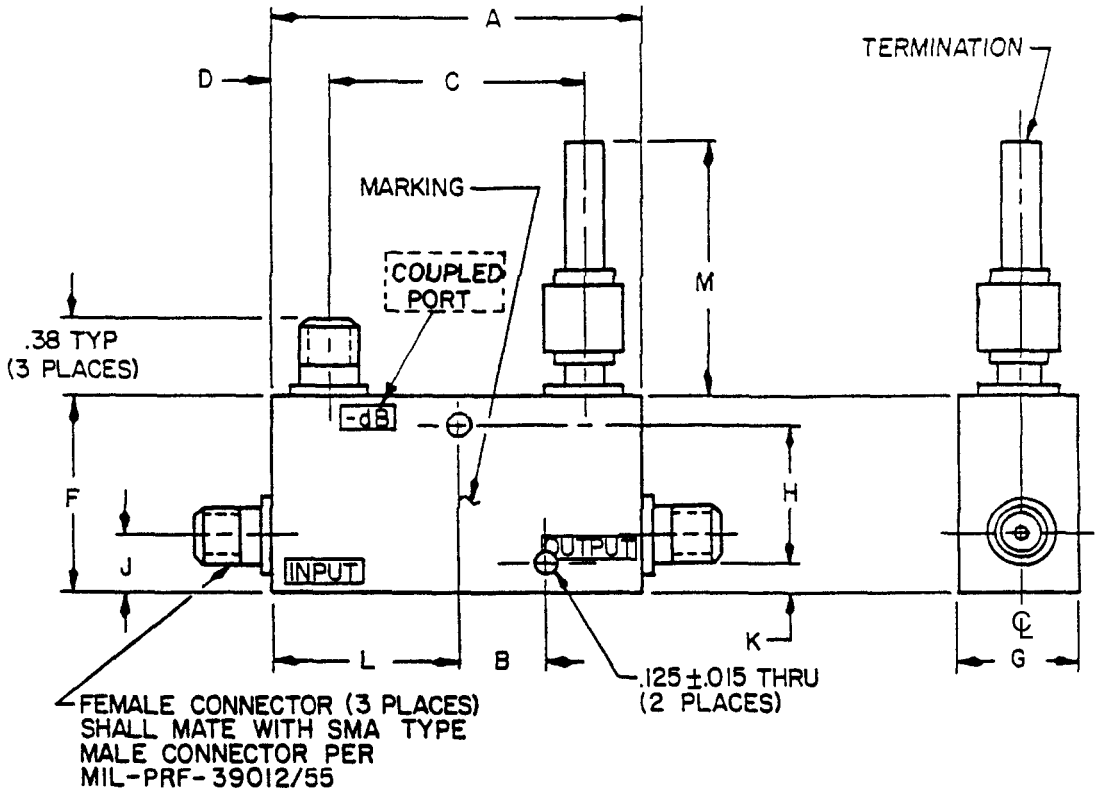


Inches	mm
.010	0.25
.093	2.36

FIGURE 1. Dimensions and configuration for part numbers M15370/10-001 through M15370/10-011, and M15370/10-019.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

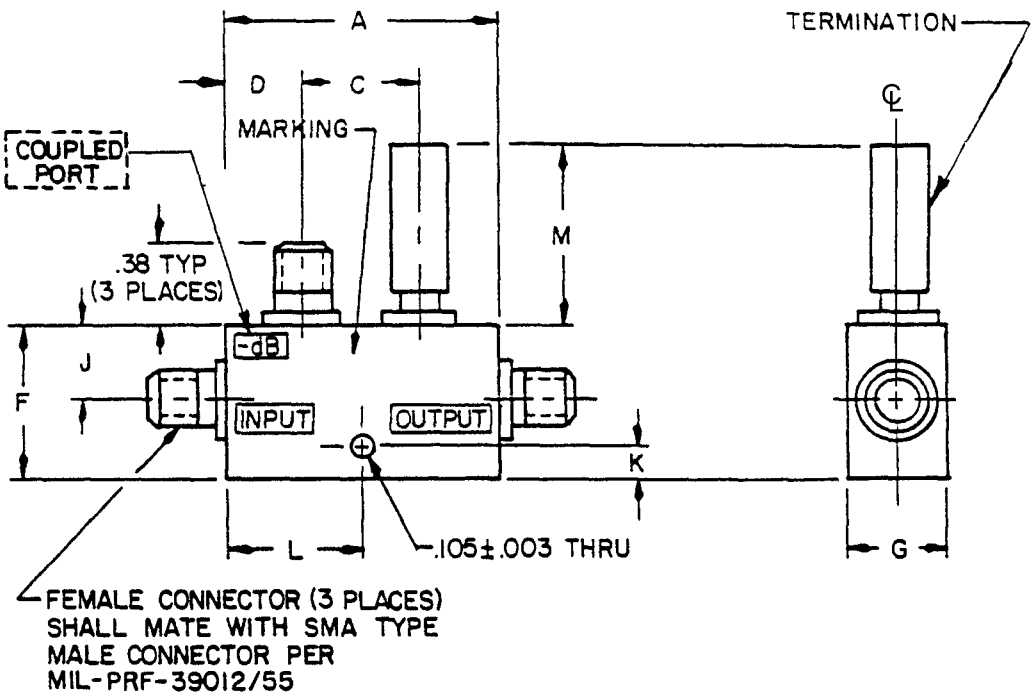
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Inches	mm
.015	0.38
.125	3.18
.38	9.65

FIGURE 2. Dimensions and configuration for part number M15370/10-012.

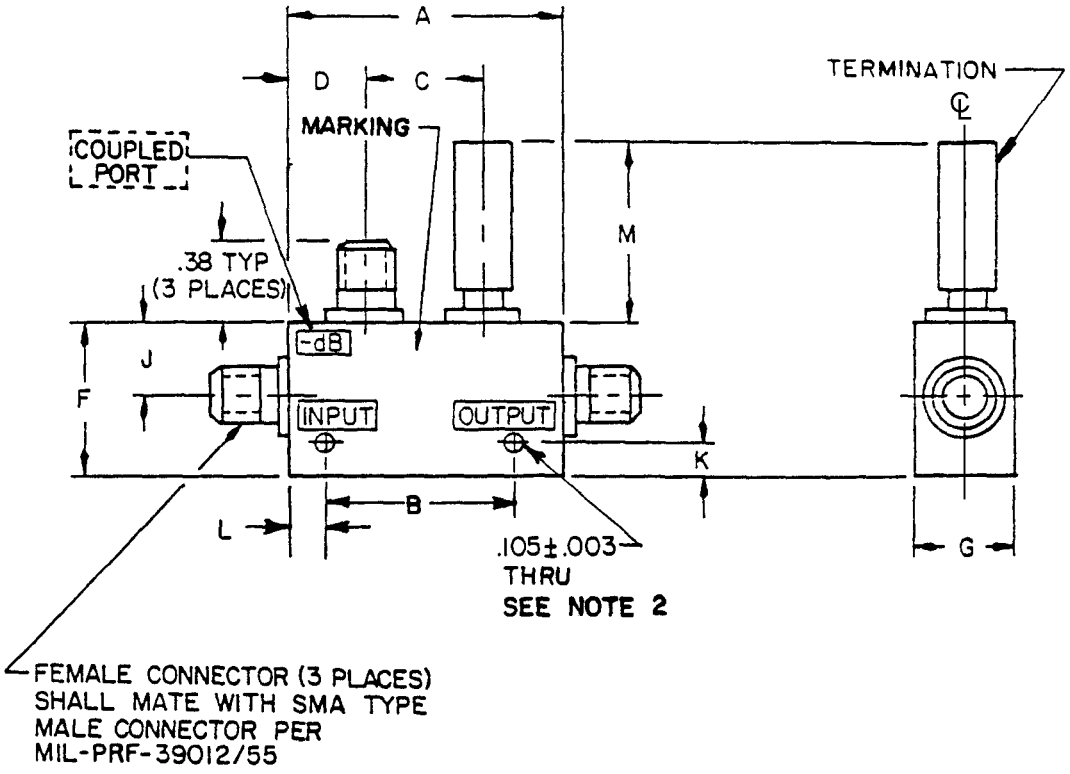
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Inches	mm
.003	0.08
.105	2.67
.38	9.65

FIGURE 3. Dimensions and configuration for part numbers M15370/10-013, M15370/10-021, M15370/10-023, and M15370/10-024.

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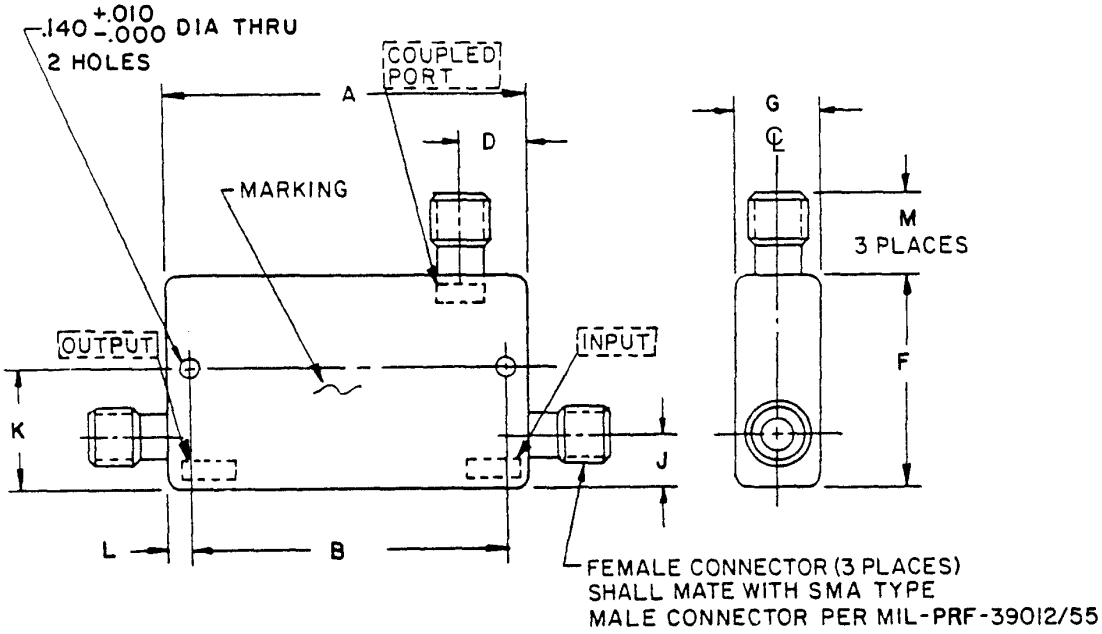


Inches	mm
.003	0.08
.105	2.67
.38	9.7

NOTES:
1. Sealant thickness, .031 maximum for dash numbers M15370/10-016, M15370/10-017, and M15370/10-018.
2. Four mounting holes are permitted, providing they are symmetrical with the two holes shown.

FIGURE 4. Dimensions and configuration for part numbers M15370/10-014 through M15370/10-018, and M15370/10-020.

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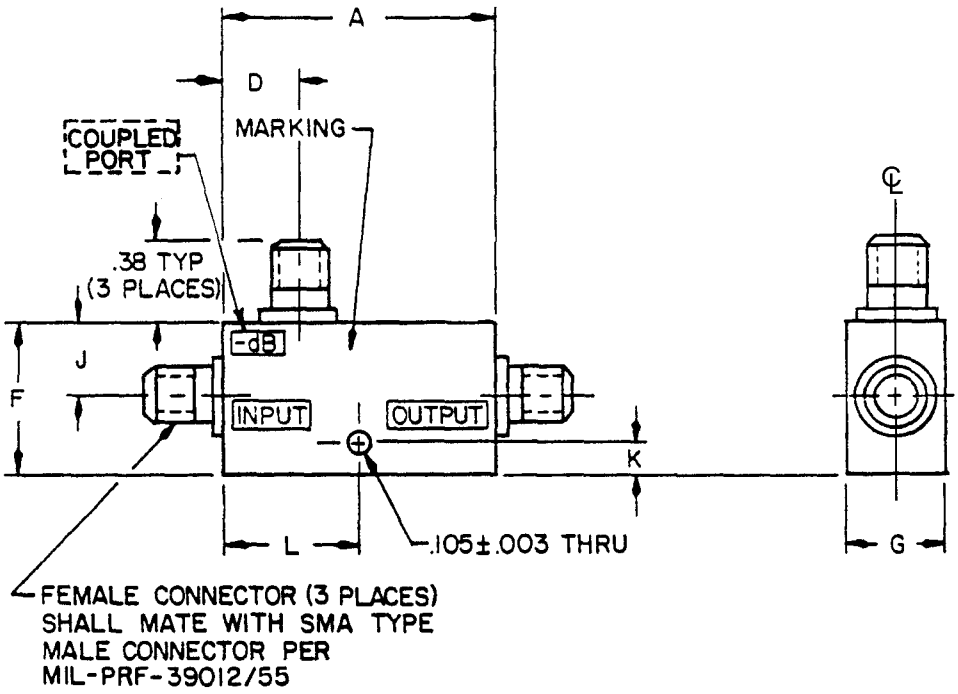


Inches	mm
.010	.25
.140	3.56

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.

FIGURE 5. Dimensions and configuration for part numbers M15370/10-022, M15370/10-025, M15370/10-026, and M15370/10-028 through M15370/10-030.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:10Z
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Inches	mm
.003	0.08
.105	2.67
.38	9.7

FIGURE 6. Dimensions and configuration for part number M15370/10-027.

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TABLE I. Dimensions and physical requirements.

Dash No.	Mounting config-uration (figure)	Dimensions 1/ 2/ 3/												Weight (max)		
		A (max)	B ±.005 (0.13)	C ±.02 (0.51)	D ±.020	E ±.02 (0.51)	F (max)	G (max)	H ±.02 (0.51)	J (max)	K +.010 -.000	L ±.010	M (max)	lbs	g	oz
001NS	1	3.320 (84.33)	3.130 (79.50)	2.710 (68.83)	.290 (7.37)	1.495 (37.97)	.530 (13.46)	.405 (10.29)	.580 (14.73)	---	---	.080 (2.03)	---	.16	71	2.5
002NS	1	3.320 (84.33)	3.130 (79.50)	2.710 (68.83)	.290 (7.37)	1.495 (37.97)	.530 (13.46)	.405 (10.29)	.580 (14.73)	---	---	.080 (2.03)	---	.16	71	2.5
003NS	1	3.320 (84.33)	3.130 (79.50)	2.710 (68.83)	.290 (7.37)	1.495 (37.97)	.530 (13.46)	.405 (10.29)	.580 (14.73)	---	---	.080 (2.03)	---	.16	71	2.5
004NS	1	2.563 (65.10)	2.391 (60.73)	1.954 (49.63)	.290 (7.37)	1.495 (37.97)	.530 (13.46)	.405 (10.29)	.580 (14.73)	---	---	.080 (2.03)	---	.16	71	2.5
005NS	1	2.563 (65.10)	2.391 (60.73)	1.954 (49.63)	.290 (7.37)	1.495 (37.97)	.530 (13.46)	.405 (10.29)	.580 (14.73)	---	---	.080 (2.03)	---	.16	71	2.5
006NS	1	2.563 (65.10)	2.391 (60.73)	1.954 (49.63)	.290 (7.37)	1.495 (37.97)	.530 (13.46)	.405 (10.29)	.580 (14.73)	---	---	.080 (2.03)	---	.16	71	2.5
007NS	1	1.715 (43.56)	1.525 (38.74)	1.105 (28.07)	.290 (7.37)	1.495 (37.97)	.530 (13.46)	.405 (10.29)	.580 (14.73)	---	---	.080 (2.03)	---	.16	71	2.5
008NS	1	1.210 (30.73)	1.020 (25.91)	.600 (15.24)	.290 (7.37)	1.495 (37.97)	.530 (13.46)	.405 (10.29)	.580 (14.73)	---	---	.080 (2.03)	---	.06	29	1.02
009NS	1	1.100 (27.94)	.920 (23.37)	.500 (12.70)	.290 (7.37)	1.620 (41.15)	.655 (16.64)	.405 (10.29)	.642 (16.31)	---	---	.080 (2.03)	---	.06	29	1.02
010NS	1	1.100 (27.94)	.920 (23.37)	.500 (12.70)	.290 (7.37)	1.620 (41.15)	.655 (16.64)	.405 (10.29)	.642 (16.31)	---	---	.080 (2.03)	---	.06	29	1.02
011NS	1	1.100 (27.94)	.920 (23.37)	.500 (12.70)	.290 (7.37)	1.620 (41.15)	.655 (16.64)	.405 (10.29)	.642 (16.31)	---	---	.080 (2.03)	---	.06	29	1.02
012NS	2	1.560 (39.62)	.375 4/ (9.53)	1.090 5/ (27.69)	.220 5/ (5.59)	---	.843 5/ (21.41)	.530 (13.46)	.605 5/ (15.37)	.281 4/ (7.14)	.130 (3.30)	.770 5/ (19.56)	1.380 (35.05)	.16	71	2.5
013NS	3	1.040 (26.42)	---	.500 5/ (12.70)	.250 5/ (6.35)	---	.843 5/ (21.41)	.415 (10.54)	---	.410 5/ (10.41)	.125 (3.18)	.521 5/ (13.23)	.800 (20.32)	.08	29	1.02
014NS	4	3.20 (81.3)	1.500 (38.10)	2.700 (68.58)	.270 (6.86)	1.45 (36.8)	.800 (20.32)	.415 (10.54)	---	.585 (14.85)	.080 (2.03)	.800 (20.32)	.650 (16.5)	.46	207	7.3
015NS	4	3.030 (76.96)	1.000 (25.40)	2.500 (63.50)	.270 (6.86)	1.00 (25.4)	.750 (19.05)	.415 (10.54)	---	.440 (11.17)	.110 (2.79)	1.013 (25.73)	.250 (6.35)	.09	43	1.5
016NS	4	3.050 (77.47)	1.500 (38.10)	2.520 (64.01)	.270 (6.86)	1.00 (25.4)	.590 (14.00)	.415 (10.54)	---	.340 (8.64)	.110 (2.79)	.790 (20.06)	.470 (11.93)	.09	43	1.5
017NS	4	3.050 (77.47)	1.750 (44.45)	2.500 (63.50)	.270 (6.86)	1.00 (25.4)	.530 (13.46)	.415 (10.54)	---	.300 (7.62)	.110 (2.79)	.650 (16.51)	.470 (11.93)	.09	43	1.5
018NS	4	3.050 (77.47)	1.750 (44.45)	2.500 (63.50)	.270 (6.86)	1.00 (25.4)	.530 (13.46)	.415 (10.54)	---	.300 (7.62)	.110 (2.79)	.650 (16.51)	.470 (11.93)	.09	43	1.5
019NS	1	2.651 (67.34)	2.391 (60.73)	1.969 (50.01)	.385 (9.78)	1.628 (41.35)	.530 (13.46)	.415 (10.54)	.578 (14.68)	---	---	.125 (3.18)	---	.09	43	1.5
020NS	4	1.200 (30.48)	.340 (8.64)	.660 (16.76)	.270 (6.86)	1.45 (36.8)	.800 (20.32)	.415 (10.54)	---	.585 (14.85)	.080 (2.03)	.410 (10.41)	.650 (16.51)	.46	207	7.3

See footnotes at end of table.

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TABLE I. Dimensions and physical requirements - Continued.

Dash number	Mounting configuration (figure)	Dimensions 1/ 2/ 3/												Weight (max)		
		A (max)	B ±.005 (0.13)	C ±.02 (0.51)	D ±.020	E ±.02 (0.51)	F (max)	G (max)	H ±.02 (0.51)	J (max)	K +.010 -.000	L ±.010	M (max)	lbs	g	oz
021NS	3	1.04 (26.4)	---	.50 (12.7)	.25 (6.4)	---	.54 (13.7)	.44 (11.9)	---	.30 (7.6)	.110 (2.79)	.520 (13.20)	.75 (19.1)	.04	17	0.60
022NS	5	2.02 (51.3)	1.75 (44.5)	---	.30 (7.6)	---	1.27 (32.3)	.44 (11.2)	---	.30 (7.6)	.625 (15.87)	.125 (3.18)	.35 (8.9)	.09	43	1.5
023NS	3	1.04 (26.4)	---	.50 (12.7)	.25 (6.4)	---	.50 (12.7)	.395 (10.0)	---	.30 (7.6)	.093 (2.36)	.507 (12.87)	.06 (1.5)	.19	85	3.0
024NS	3	1.04 (26.4)	---	.50 (12.7)	.25 (6.4)	---	.54 (13.7)	.395 (10.0)	---	.30 (7.6)	.093 (2.36)	.521 (13.23)	.812 (20.62)	.04	17	0.60
025NS	5	2.02 (51.3)	1.75 (44.5)	---	.25 (6.4)	2.0 (51.)	1.6 (41.)	.395 (10.0)	---	.30 (7.6)	.625 (15.87)	.125 (3.18)	.40 (10.2)	.09	43	1.5
026NS	5	2.02 (51.3)	1.75 (44.5)	---	.25 (6.4)	2.0 (51.)	1.6 (41.)	.395 (10.0)	---	.30 (7.6)	.625 (15.87)	.125 (3.18)	.40 (10.2)	.09	43	1.5
027NS	6	1.04 (25.4)	---	---	.250 (6.35)	---	.50 (12.7)	.395 (10.0)	---	.30 (7.6)	.093 (2.36)	,500 (12.70)	---	.06	28	1.0
028NS	5	2.02 (51.3)	1.75 (44.5)	---	.35 (8.9)	---	1.27 (32.3)	.44 (11.2)	---	.30 (7.6)	.625 (15.87)	.125 (3.18)	.35 (8.9)	.09	43	1.5
029NS	5	2.02 (51.3)	1.75 (44.5)	---	.30 (7.6)	---	1.27 (32.3)	.44 (11.2)	---	.30 (7.6)	.625 (15.87)	.125 (3.18)	.35 (8.9)	.09	43	1.5
030NS	5	2.02 (51.3)	1.75 (44.5)	---	.30 (7.6)	---	1.27 (32.3)	.44 (11.2)	---	.30 (7.6)	.625 (15.87)	.125 (3.18)	.35 (8.9)	.09	43	1.5

1/ Dimensions are in inches.
2/ Metric equivalents are given for general information only.
3/ Metric equivalents are in parentheses.
4/ Tolerance is ± .010 (0.25).
5/ Tolerance is ± .030 (0.76).

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TABLE II. Electrical requirements.

Dash number	Frequency range	Coupling (dB)		Coupling variation (dB) max	Effective directivity (dB) min	Insertion loss (dB) max	VSWR max		RF input power to primary			Reflected power (avg) max
		Nominal	Tolerance				Primary line	Secondary line	Over frequency range	Avg	Peak	
	GHz								Watts	Watts	KW	Watts
001NS	0.5 - 1.0	6	±1.0	±0.6	20	0.2	1.25:1	1.25:1	50	50	0.25	2
002NS	0.5 - 1.0	10	±1.0	±0.75	20	0.2	1.25:1	1.25:1	50	50	0.25	5
003NS	0.5 - 1.0	20	±1.0	±1.0	20	0.2	1.25:1	1.25:1	50	50	0.25	50
004NS	1.0 - 2.0	6	±1.0	±0.5	20	0.2	1.25:1	1.25:1	50	50	0.25	2
005NS	1.0 - 2.0	10	±1.0	±0.75	20	0.2	1.25:1	1.25:1	50	50	0.25	5
006NS	1.0 - 2.0	20	±1.0	±1.0	20	0.2	1.20:1	1.20:1	50	50	0.25	50
007NS	2.0 - 4.0	6	±1.0	±0.5	18	0.2	1.20:1	1.20:1	50	50	0.25	2
008NS	4.0 - 8.0	6	±1.0	±0.5	18	0.2	1.20:1	1.20:1	50	50	0.25	2
009NS	8.0 - 12.4	10	±1.0	±0.5	15	0.25	1.30:1	1.30:1	50	50	0.25	2
010NS	8.0 - 12.4	20	±1.0	±0.75	15	0.25	1.30:1	1.30:1	50	50	0.25	2
011NS	12.4 - 18.0	6	±1.0	±0.5	12	0.25	1.35:1	1.35:1	50	50	0.25	2
012NS	7.0 - 11.0	10	±1.0	±0.75	15	0.85	1.25:1	1.25:1	100	100	2.0	10
013NS	7.0 - 11.0	20	±1.0	±1.0	15	0.40	1.25:1	1.30:1	100	100	2.0	100
014N 1/ S	0.5 - 1.0	10	±1.25	±0.75	25	0.20	1.10:1	1.10:1	50	50	3.0	5
015NS	0.5 - 1.0	30	±1.0	±0.75	25	0.15	1.10:1	1.10:1	50	50	4.0	50
016NS	0.5 - 1.0	10	±1.0	±0.75	25	0.2	1.15:1	1.15:1	50	50	3.0	5
017NS	0.5 - 1.0	20	±1.0	±0.75	25	0.2	1.15:1	1.15:1	50	50	3.0	50
018NS	0.5 - 1.0	30	±1.0	±0.75	25	0.2	1.15:1	1.15:1	50	50	3.0	50
019NS	1.5 - 1.9	15	±1.0	±0.75	20	0.2	1.20:1	1.20:1	5	5	2.0	2
020N 2/ S	2.0 - 4.0	6	±1.0	±0.5	20	0.2	1.30:1	1.30:1	50	50	2.0	2

See footnotes at end of table.

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TABLE II Electrical requirements.

Dash number	Frequency range	Coupling (dB)		Coupling variation (dB) max	Effective directivity (dB) min	Insertion loss (dB) max	VSWR max		RF input power to primary			Reflected power (avg) max
		Nominal	Tolerance				Primary line	Secondary line	Over frequency range	Avg	Peak	
	GHz								Watts	Watts	KW	Watts
021NS	2.6 - 5.2	20	±1.25	±0.75	20	0.25	1.25:1	1.25:1	50	50	3.0	50
022NS 3/	2.6 - 5.2	20	±1.25	±0.75	20	0.25	1.25:1	1.25:1	50	50	7.5	50
023NS	4.4 - 5.0	10	±0.45	±0.2	20	0.2	1.25:1	1.25:1	20	20	2	1
024NS	7.0 - 11.0	6	±1.0	±0.6	17	0.5	1.30:1	1.30:1	50	50	3	2
025NS	8.75 - 9.65	6	±0.5	±0.3	17	0.3	1.20:1	1.20:1	1	1	1	2
026NS	8.75 - 9.65	30	±1.0	±0.5	17	0.3	1.20:1	1.20:1	1	1	1	50
027NS	9.0 - 9.2	20	±0.6	±0.3	18	0.4	1.35:1	1.35:1	50	50	1	20
028NS 4/	8.0 - 12.4	10	±1.0	±0.5	15	0.3	1.30:1	1.30:1	5	5	5	2
029NS	8.0 - 12.4	25	±0.5	±0.3	20	0.3	1.25:1	1.25:1	35	35	5	50
030NS	8.0 - 12.4	30	±1.0	±0.5	20	0.3	1.25:1	1.25:1	35	5	5	50

1/ Inactive for new design, use dash number -002 for new design.
2/ Inactive for new design, use dash number -007 for new design.
3/ Inactive for new design, use dash number -021 for new design.
4/ Inactive for new design, use dash number -009 for new design.

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REQUIREMENTS:

Design, construction, and physical dimensions: See figures 1 through 6 and table I.

Operating temperature range:

-54°C to +85°C for dash numbers 022 through 028.
-55°C to +100°C for dash numbers 014 and 020.
-54°C to +110°C for dash numbers 001 through 013, 015 through 019, 021, 023 through 027, 029, and 030.

Weight: See table I.

Electrical: See table II.

Marking: Part number (see tables I and II) shall be located as shown on figures 1 through 6.

Part number: M15370/10- (and a dash number from table I).

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Navy - AS, CG, MC, OS, SH
Air Force - 99

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